Assignment-based Subjective Questions

Question 1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable? (Do not edit)

Total Marks: 3 marks (Do not edit)

Answer: <Your answer for Question 1 goes below this line> (Do not edit)

From the analysis of categorical variables such as season, weathersit, yr, holiday, and workingday, it is evident that they have a significant impact on bike demand. Bike rentals are generally higher during summer and fall seasons compared to spring and winter. Favorable weather conditions like clear or misty weather result in higher demand, whereas rainy or snowy conditions reduce bike usage. The variable yr shows that demand increased in 2019 compared to 2018, indicating growing popularity of bike-sharing services. Additionally, working days tend to have higher demand than holidays, suggesting regular commuting behavior contributes significantly to rentals.

Question 2. Why is it important to use **drop_first=True** during dummy variable creation? (Do not edit)

Total Marks: 2 marks (Do not edit)

Answer: <Your answer for Question 2 goes below this line> (Do not edit)

Using drop_first=True helps prevent the dummy variable trap, which occurs due to multicollinearity when all categories of a categorical variable are included. Since the dropped category can be inferred from the remaining dummy variables, keeping all of them would make the features linearly dependent. Removing one category ensures model stability and allows linear regression to estimate coefficients correctly.

Question 3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable? (Do not edit)

Total Marks: 1 mark (Do not edit)

Answer: <Your answer for Question 3 goes below this line> (Do not edit)

Among the numerical variables, temperature (temp) shows the highest positive correlation with the target variable cnt. This indicates that bike demand increases significantly as temperature becomes more favourable.

Question 4. How did you validate the assumptions of Linear Regression after building the model on the training set? (Do not edit)

Total Marks: 3 marks (Do not edit)

Answer: <Your answer for Question 4 goes below this line> (Do not edit)

The assumptions of linear regression were validated using multiple techniques:

- **Linearity** was checked using scatter plots between predicted values and residuals.
- **Normality of errors** was assessed using a histogram and density plot of residuals.
- **Homoscedasticity** was verified by plotting residuals against predicted values to ensure constant variance.

- **Multicollinearity** was addressed by removing correlated features and using VIF analysis. These checks confirmed that the linear regression assumptions were reasonably satisfied.

Question 5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes? (Do not edit)
Total Marks: 2 marks (Do not edit)
Answer: <Your answer for Question 5 goes below this line> (Do not edit)
The top three features contributing significantly to bike demand are: Temperature (temp), Year (yr), Weather situation (clear vs adverse weather)

These variables strongly influence user behaviour and explain a major portion of the variability in bike demand.

General Subjective Questions

Question 6. Explain the linear regression algorithm in detail. (Do not edit)
Total Marks: 4 marks (Do not edit)
Answer: Please write your answer below this line. (Do not edit)

Linear regression is a supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. The algorithm estimates coefficients by minimising the sum of squared residuals (least squares method). Linear regression is widely used due to its simplicity, interpretability, and effectiveness when assumptions are met.

Question 7. Explain the Anscombe's quartet in detail. (Do not edit)
Total Marks: 3 marks (Do not edit)
Answer: Please write your answer below this line. (Do not edit)

Anscombe's quartet consists of four datasets that have nearly identical statistical properties such as mean, variance, correlation, and regression line, yet differ significantly when visualised. This demonstrates the importance of data visualisation, as relying solely on summary statistics can be misleading. The quartet highlights that visual inspection is crucial to detect non-linearity, outliers, or unusual distributions before applying statistical models.

Question 8. What is Pearson's R? (Do not edit)
Total Marks: 3 marks (Do not edit)
Answer: Please write your answer below this line. (Do not edit)

Pearson's R is a statistical measure that quantifies the strength and direction of the linear relationship between two continuous variables. Its value ranges from –1 to +1, where:

- +1 indicates perfect positive correlation,
- –1 indicates perfect negative correlation,
- 0 indicates no linear correlation.

Question 9. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling? (Do not edit)
Total Marks: 3 marks (Do not edit)
Answer: Please write your answer below this line. (Do not edit)

Scaling is the process of bringing features onto a **similar scale** so that no single feature dominates model training. It is especially important for algorithms sensitive to feature magnitude.

- Normalization (Min-Max Scaling) rescales values to a fixed range, usually [0,1].
- Standardization (Z-score Scaling) transforms data to have mean 0 and standard deviation 1.

Standardization is preferred for linear regression when features follow a Gaussian distribution.

Question 10. You might have observed that sometimes the value of VIF is infinite. Why does this happen? (Do not edit)
Total Marks: 3 marks (Do not edit)
Answer: Please write your answer below this line. (Do not edit)

An infinite VIF value occurs when a feature is perfectly multicollinear, meaning it can be exactly predicted from other features. This usually happens due to redundant variables or incorrect dummy variable encoding. In such cases, regression coefficients become unstable, and the feature should be removed.

Question 11. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression. (Do not edit)
Total Marks: 3 marks (Do not edit)
Answer: Please write your answer below this line. (Do not edit)

A Q-Q (Quantile–Quantile) plot compares the distribution of model residuals with a theoretical normal distribution. If residuals lie along the diagonal line, it indicates normality. Q-Q plots are important in linear regression to verify the normality of errors, which is a key assumption for valid inference and hypothesis testing.
